INSTACART

Instacart Project Roadmap

Version 2.0

The next milestones after Python automation

Current status, practical boundaries and evidence-based completion

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Verified through 31 October 2026
Code checkpoint: `bfc1763`

Next: continuous integration	Reuse uv + unittest	Cloud/dbt branch planned
ML extension planned	Runtime gates explicit	Deployment optional

Completed work and planned extensions are clearly separated. Future milestones become complete only after their acceptance evidence is recorded.

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What changed from the first roadmap

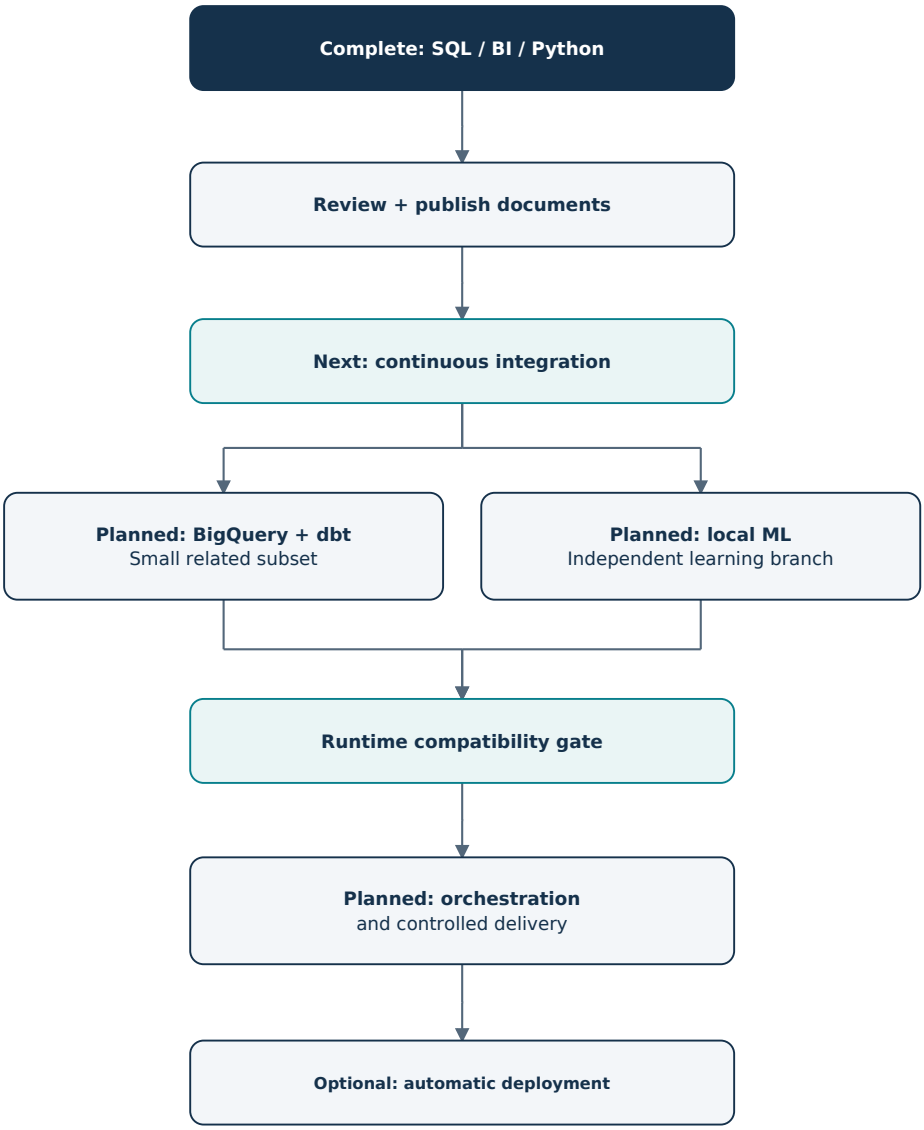
The earlier draft dated 29 October treated Python automation as future work. The local five-file pipeline is now implemented and verified. This revision records that result, retains the original stage numbers for continuity, and gives the remaining stages practical completion checks.

The first CI phase now reuses the existing `unittest` suite and uv environment. Cloud/dbt, ML, orchestration, and delivery remain future work. Windows-to-Linux database connectivity is an explicit feasibility gate before a containerized deployment is promised.

1. Current position and remaining work

Original stage	Milestone	State on 31 October	Evidence or next action
1	SQL and local Power BI analysis	Complete	Scripts 00-21, three views, seven measures, three pages
2	GitHub project delivery	Core files pushed; smoke check open	Verify a clean download/open of the LFS-tracked PBIX
3	Python loading, checks, and run reporting	Core local milestone complete	`bfc1763`, 53 tests, safe replay and recovery evidence
Documentation	Python report, scope, roadmap	Created for review/local saving	Remote publication not yet verified
4	Continuous integration	Planned - next implementation	Automate the existing deterministic test suite
5	BigQuery and dbt	Planned	Small related subset, cloud queries, models/tests
6	Reorder prediction	Planned	Labels, history-only features, baseline/model evaluation
7	Airflow orchestration	Planned with environment gate	Compatible tasks, dependency graph, recovery proof
8	Continuous delivery	Planned with runtime gate	Traceable runnable release artifact and smoke test
9	Automatic deployment	Optional; target undecided	Separate target, cost/authentication, and recovery decision
10	Portfolio / LinkedIn	Separate discussion	Draft only when Saurabh initiates that conversation

No calendar deadline is attached. Effort depends on tool compatibility and access. The stages represent evidence to produce, not an obligation to add every technology before the project is useful.



Milestone dependencies and conditional branches

The cloud/dbt branch and local ML branch share the validated data foundation but do not require each other to run. We will normally complete CI first, then a small cloud/dbt demonstration, then ML. Airflow coordinates suitable tasks after they work individually; release packaging follows runtime feasibility.

2. Starting baseline and completion rules

2.1 Results to preserve

The descriptive baseline includes 3,214,874 prior orders, 32,434,489 entries, 19,126,536 reordered entries, and 49,688 products. The new Python tables include all 3,421,083 order metadata rows. Keep the order-cohort distinction visible in every comparison.

The completed local command is:

```
uv run python -m scripts.run_pipeline --data-dir "D:\DataProjects\data-download"
```

It verifies baseline bytes, runs five table loaders in dependency order, and checks committed automation tables. Each loader has its own transaction. The original reporting-view runner remains separate:

```
uv run python -m scripts.run_validations
```

2.2 Status meanings

Label	Meaning
Complete	Implementation plus stated acceptance evidence exists
Planned	Agreed learning direction; implementation has not been established
Conditional	Start only after access, compatibility, resource, or target feasibility is resolved
Open check	Narrow verification still missing from the record
Created for review	Document prepared; review/commit/publication are separate actions

For every future phase: implement, run relevant checks, deliberately exercise a meaningful failure where appropriate, preserve evidence, document limits, then update the status. Do not keep adding tools to avoid closing a milestone.

3. Immediate documentation and delivery checkpoint

Before starting new implementation, save the completed Python report, this roadmap, and the scope document under `docs`. Preserve their relative asset folders so diagrams render in GitHub. Review historical statements and proposed outcomes, then commit the accepted documents.

The older input runbook still describes the loader as its next step, and the README primarily introduces the SQL/Power BI stage. A small documentation follow-up should add navigation and current status without rewriting the already validated analytical definitions.

Check	Completion evidence
Python report saved	Markdown/PDF and eight diagram assets available in local docs
Scope and roadmap accepted	Current statuses, exclusions, and phase gates reviewed
Documentation publication	Commit and remote branch state verified
PBIX delivery	Fresh LFS-enabled clone/download obtains the actual binary, opens in Desktop, and can reconnect to the local SQL source

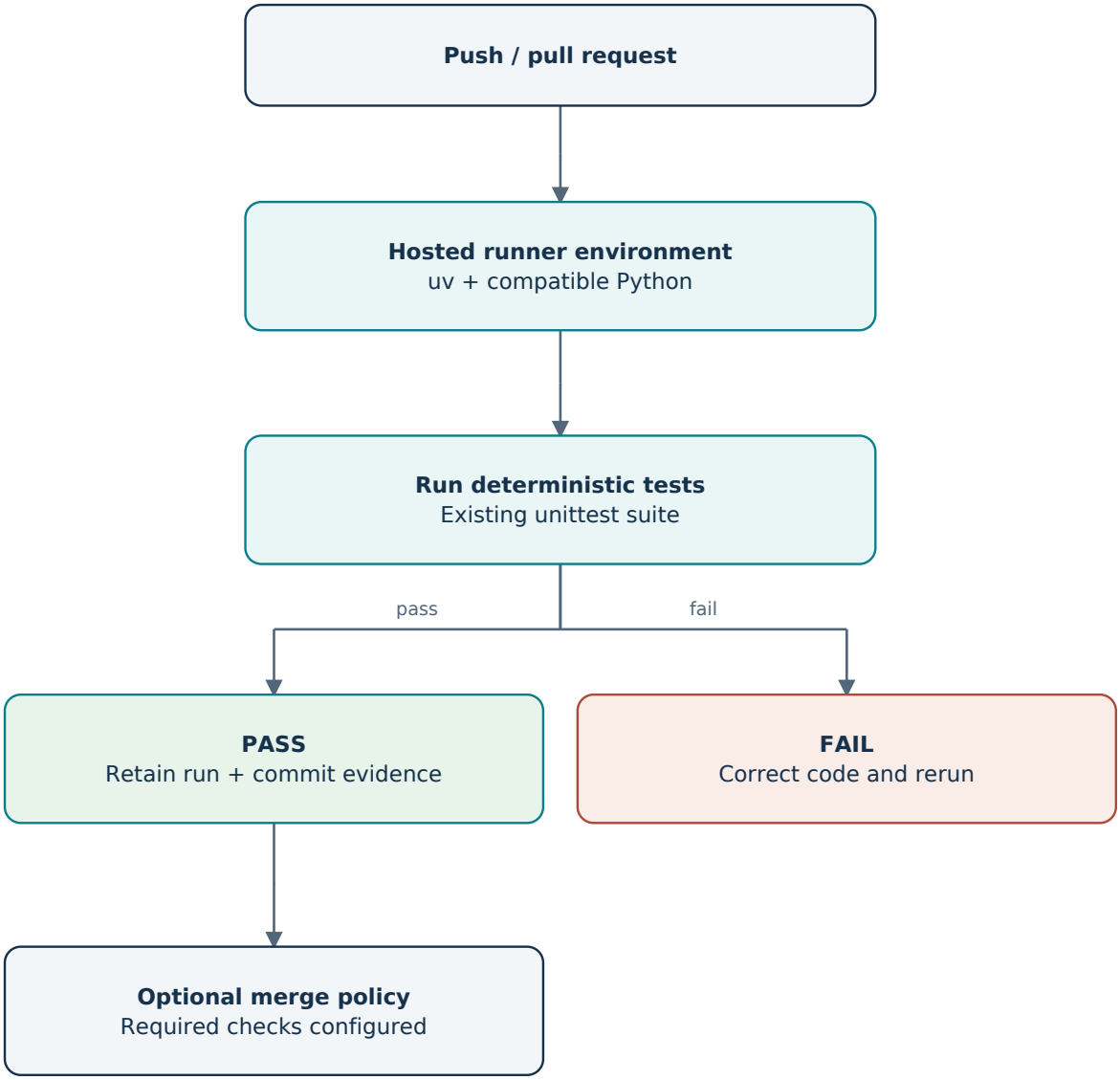
Downloading an LFS pointer alone is insufficient for the PBIX check. The smoke test is a delivery verification, not a reason to rebuild every chart or repeat the full analytical study.

4. Stage 4 - Continuous integration

4.1 Explain it simply

Continuous integration means code changes trigger automatic checks in a clean runner environment. The code has already been tested manually on the laptop. CI repeats the relevant code checks on pushes/pull requests so later changes are less likely to break accepted behavior [R1].

Continuous delivery means a tested release artifact can be prepared and used. Continuous deployment goes further and updates a running target automatically. The first next milestone is CI; it does not require either deployment form.



Proposed CI check and release gate

4.2 Technology and scope

Use GitHub Actions, uv, Python 3.14, and `unittest`. Begin with a hosted Windows runner to match the development environment. Validate the runner label and action versions at implementation time; pin reviewed action versions/commits rather than copying an old example blindly. Add a Linux matrix only if it serves a tested portability goal.

The first job uses deterministic small fixtures, mocks, and the small subprocess test. It does not connect to Saurabh's laptop SQL Server or load the original 32-million-row table. Hosted runners are their own

environments [R2]. A database integration job can later create its own isolated test database and fixtures.

4.3 Proposed workflow behavior

Workflow step	Purpose
Checkout code	Test the exact commit under review
Prepare uv and compatible Python	Recreate the declared environment
`uv sync --locked`	Require dependency metadata and lockfile agreement
`uv run --locked python -m unittest discover -s tests -v`	Run the current suite with visible failures
Retain useful test output	Link future failures to a run/commit
Finish nonzero on test failure	Make the workflow status meaningful

Proposed file: `.github/workflows/python-tests.yml`. This is a planned path, not an existing workflow claim. Use the permission needed to read/test code; credentials, real raw files, and the PBIX binary are unnecessary inputs to this first job.

4.4 Acceptance and failure exercise

1. A valid push/pull request starts the workflow and passes the relevant suite.
2. A controlled change on a temporary branch makes one test fail and the workflow becomes red.
3. Correcting that change makes the next run green.
4. Save workflow run links, source commits, and the failure/correction explanation.
5. If merge blocking is desired, verify the repository's available branch/ruleset support and configure required checks separately. A workflow alone does not enforce that policy.

Done when: the checks run automatically and the controlled fail/correct/pass sequence is inspectable. Reusing the 53 tests is the starting point; retain the correct evolving suite, not an arbitrary permanent count.

5. Stage 5 - Small BigQuery and dbt demonstration

5.1 Why add this branch

The local project demonstrates SQL Server work. A cloud branch can show that the same data questions and quality rules can be expressed in another analytical warehouse. dbt adds named SQL models, dependencies, tests, and documentation rather than a collection of disconnected query tabs.

5.2 Subset before platform setup

Choose a reproducible customer subset, initially around 5,000 users if its measured size is comfortable. Preserve complete relevant prior histories, related orders, and required product/aisle/department records. The exact customer rule, counts, and fingerprints become a subset manifest.

Do not use the first N rows of each CSV independently: that can discard related records and make local/cloud comparisons meaningless. Reconcile exactly the same selected subset in both systems. Reuse SQL Server aggregation to keep Python memory use small.

5.3 Tools and checks

Component	Proposed use
BigQuery sandbox	Batch-load the small subset and run analytical SQL
dbt with compatible BigQuery adapter	Build staging/reporting models and declare dependencies
dbt generic tests	Unique keys, required IDs, accepted values, relationships
Custom SQL test	Composite order-product uniqueness and project-specific gap/total checks
Local SQL Server	Produce matching reference results on that same subset

The sandbox currently limits capabilities, including DML and streaming, and expires stored tables/views after 60 days [R3]. Keep this branch small and reproducible; recheck quota/authentication details at setup. Use suitable batch, view, or full-rebuild models instead of assuming incremental MERGE models will work. First prove a tiny end-to-end adapter model before promising the full dbt setup.

dbt's generic tests include `unique`, `not_null`, `accepted_values`, and `relationships`; custom queries can test project-specific assumptions [R4]. Version and adapter compatibility are implementation gates. A local DuckDB/dbt fallback demonstrates local dbt work, not BigQuery deployment experience.

5.4 Planned outputs and completion

Proposed folders: `cloud/bigquery/`, `dbt_instacart/`, and `docs/cloud_evidence/`. Keep secrets and machine-specific connection profiles outside committed source.

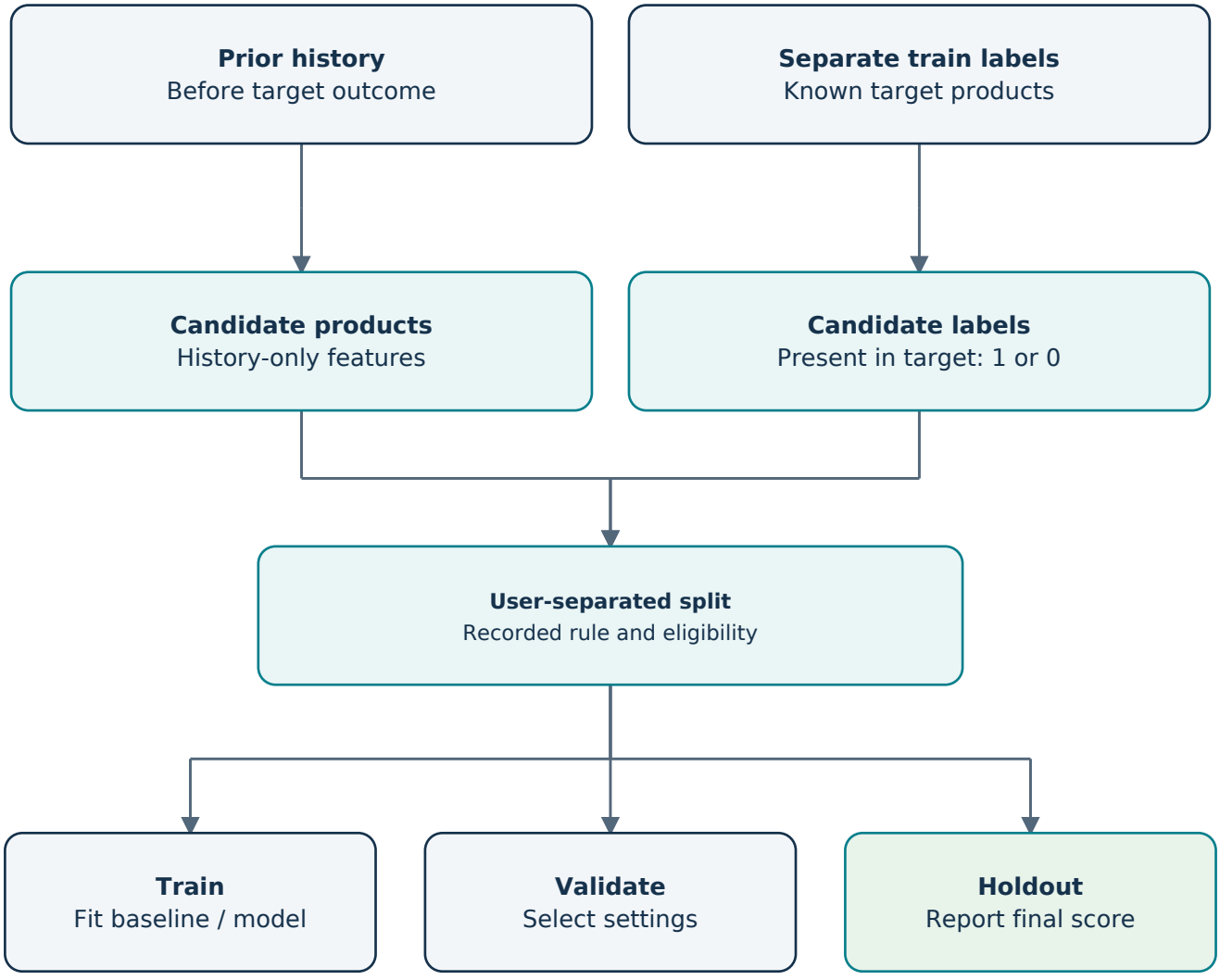
Done when: the subset can be recreated; selected product/order/reorder results reconcile across SQL Server and BigQuery; dbt builds/tests pass; one deliberately broken test condition is detected; generated model documentation and a short SQL-dialect note are preserved. No paid expansion is assumed.

6. Stage 6 - Machine-learning task and data preparation

6.1 The first prediction question

For a customer with earlier shopping history, which products they purchased before will appear in their next order? Each candidate example represents a customer, target order, and previously purchased product. Its label is 1 if that product appears in the known target order; otherwise 0.

This is an initial reordered-product task. New-product discovery, quantities, revenue, live APIs, and real-time recommendations require separate scope. The historical `reordered` flag describes a past entry; it is not automatically the correct label for a future prediction example.



History, labels, and customer-separated evaluation

6.2 Inputs and label boundary

Input	Role	Work remaining
Prior metadata and entries	Features from earlier purchases	Define exact per-target history cutoff
Train metadata and `order_products__train.csv`	Known next-order outcomes	Add a separate label manifest/contract and loader; validate references
Official test cohort	Unlabeled target orders	Optional batch prediction output after evaluation; no direct local ground-truth score

Keep the train-label file outside the accepted five-file baseline until its separately versioned contract is agreed. Adding it should not silently alter the completed baseline recorder or target counts.

6.3 Proposed split and leakage rules

Start with an explicit reproducible customer assignment, initially proposed as 70% train, 15% validation, and 15% final holdout among eligible labeled users. Record IDs, seed/rule, sizes, exclusions, and candidate coverage. Keep all candidate rows for one customer in one partition.

Every feature must use information available before that customer's target outcome. Build each customer's own features from permitted earlier history. Fit learned preprocessing and shared training statistics on the allowed training partition, then apply them to validation/holdout; do not tune on final holdout outcomes [R5].

The user-grouped split measures generalization across held-out users with their observed histories. It is not a claim of performance over future calendar dates; this dataset does not provide full transaction dates in the current scope. If later using multiple targets from one user, preserve group separation and cutoff rules explicitly.

7. Stage 6 - Features, models, evaluation, and batch prediction

7.1 Manageable feature design

Feature family	Initial examples	Computation boundary
User-product	Historical purchase count, fraction of prior orders, orders since last purchase, previous-basket presence	Earlier history only
User	Prior order count and average historical basket size	Before target order
Product	Popularity/repeat statistics	Training-side reference history under the split policy
Target context	Known ordering context, if justified and available at prediction time	Exclude target outcome and target basket contents

Use SQL Server to aggregate the large history into a smaller feature table. Load only the selected feature set into Python. Keep candidate generation, preprocessing, and labels traceable. Measure memory/runtime on the reproducible subset before scaling.

7.2 Modeling order

1. Implement a simple documented customer-frequency recommendation baseline.
2. Fit a scikit-learn logistic-regression model with suitable preprocessing.
3. Consider histogram gradient boosting only after the baseline/first model and evaluation are working.
4. Choose probability thresholds or set-selection rules using validation results.
5. Evaluate the final accepted settings once on the reserved holdout and report errors honestly.

CPU work is sufficient for the intended initial scope. The GTX 1650 is not a requirement or a reason to add a deep-learning model. Pin compatible data/ML packages in a reviewed environment; avoid expanding the ingestion environment without need.

7.3 Evaluation contract

Question	Required answer in the ML report
Does it beat a simple rule?	Baseline and model evaluated on exactly the same held-out orders
Does it over-recommend?	Precision and representative false positives
Does it miss expected products?	Recall and representative false negatives
What is the set-level quality?	Mean per-order F1 over the declared reordered-product candidate task
What happens for empty sets?	Explicit handling of empty truth/prediction sets
Is it fair?	Feature cutoffs, customer partitions, and no holdout tuning
Can it be repeated?	Dataset/feature version, split manifest, seed, model settings, commit, and run ID

The local task's F1 definition and candidate truth set must be stated. Previously purchased candidates omit new products by design, so do not automatically equate this restricted score with a full-basket competition leaderboard score. Official test predictions are an optional output, not a substitute for labeled holdout evaluation.

Done when: baseline and model results are reproducible, leakage checks pass, errors are explained, and batch predictions can be regenerated from a recorded model. A model is not required to beat the baseline for the learning exercise to be valid; the result must be reported accurately.

Proposed outputs: `ml/`, small versioned feature/split manifests, `docs/ML_REPORT.md`, and managed model/prediction artifacts. Model binaries and large feature exports need a storage decision rather than automatic ordinary-Git inclusion.

8. Stage 7 - Orchestration and recovery

8.1 What Airflow adds

The local coordinator already establishes task order. Airflow would add scheduler-managed task instances, visible states, and controlled reruns/retries for suitable commands. It should invoke accepted job logic rather than hide duplicated business rules in DAG definitions.

Before setup, decide which tasks are compatible with the runtime. Airflow uses Linux/WSL2 or Linux containers on Windows [R6]. The existing local connection uses Windows integrated authentication and local server-side bulk paths. Those assumptions do not automatically carry into a Linux worker.

Feasibility decision	Acceptable resolution
Runtime memory	A measured small learning setup fits the laptop alongside required services
Python/tool compatibility	Pin a supported Airflow environment separately from ingestion/ML if needed
SQL authentication/networking	Prove a supported explicit connection; do not assume localhost/Windows identity is shared
Bulk source path	Confirm SQL Server can read the chosen server-side path
Task scope if Windows loading remains external	Orchestrate a compatible cloud/subset/feature demo and document the Windows loader as a separate manual prerequisite

The Docker quick start calls for at least 4 GB memory and ideally more; on an approximately 12 GB laptop, use a measured small learning configuration and avoid running all heavy tools together [R7]. The quick start is a learning setup, not evidence of production availability.

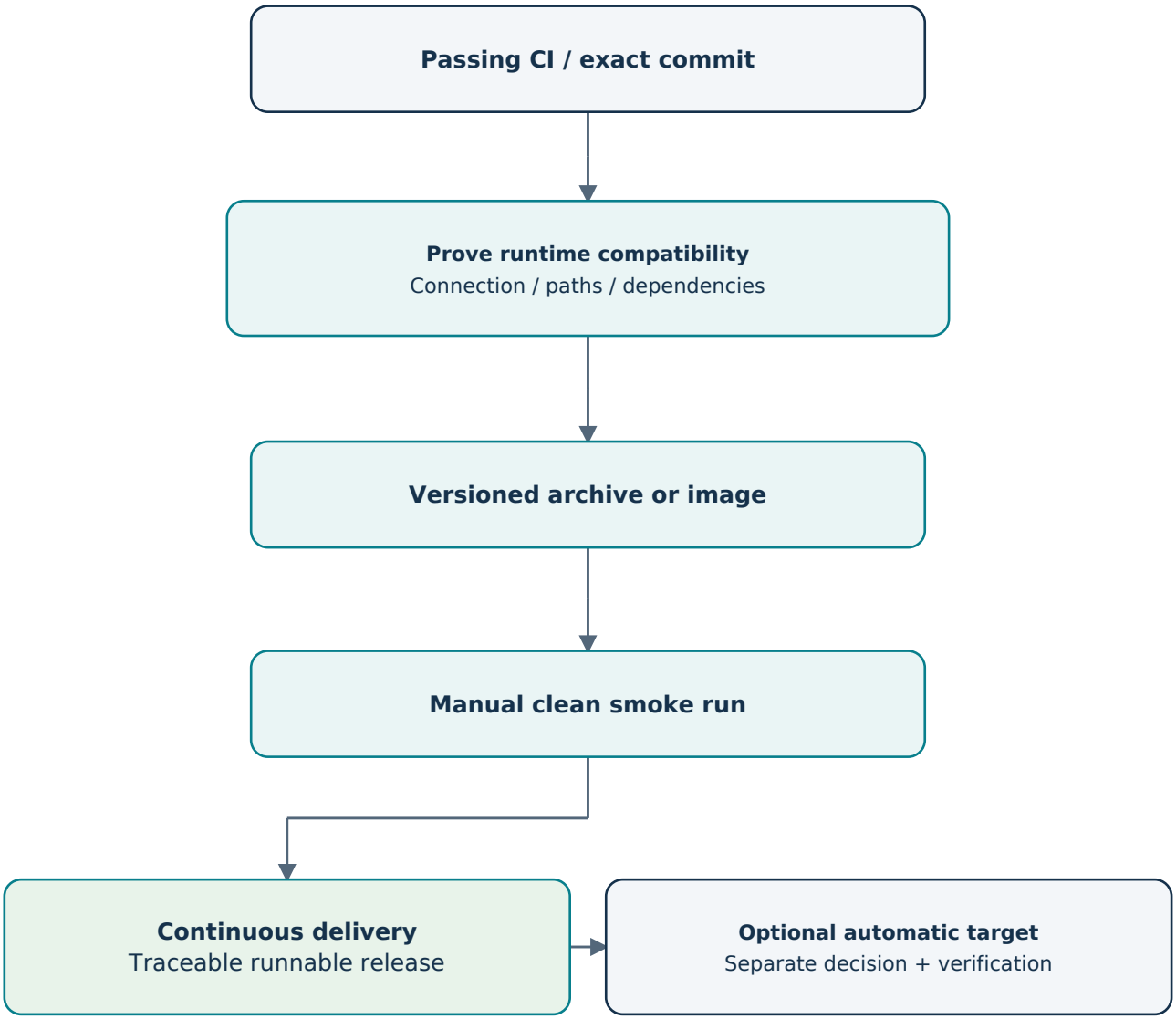
8.2 Failure behavior and acceptance

Transient connection errors can have bounded retries if rerunning the task is safe. Missing files, wrong schemas, and deterministic quality failures should stop and be corrected rather than retried indefinitely. Failed validation must prevent downstream feature publication or scoring.

Keep model training separate from routine batch scoring. Scoring uses a recorded accepted model. Retraining, drift handling, and automated model promotion remain later decisions.

Done when: an honestly scoped compatible DAG runs; one controlled failure blocks the appropriate downstream tasks; a corrected rerun succeeds without duplication; task/run states and recovery steps are preserved. Scheduling static input replay proves orchestration behavior, not a live data feed.

9. Stages 8-9 - Delivery and optional deployment



Release artifact versus a deployed running target

9.1 Stage 8: continuous delivery

Choose a release artifact that can actually run the accepted job. A versioned code/archive release is a valid first form. A Docker image published to GitHub Container Registry is a useful option after SQL access, filesystem paths, required drivers, and authentication have been proved for that image.

The image option is conditional, not a claim that copying Windows Python files into a Linux container yields a working full SQL pipeline. If only the portable subset/ML task is packaged, the release must name that scope accurately.

Build from an exact commit after passing CI; record dependency and artifact versions; exclude raw CSVs, credentials, `.venv`, routine logs, and the PBIX from the runtime package. Supply runtime configuration separately. A manual smoke run proves the artifact works in its documented environment.

Done when: passing checks produce a traceable artifact, a clean environment can execute the documented smoke workflow, and a previous artifact can be selected and checked for recovery. Retaining old artifacts is useful only if their data/model compatibility is also recorded.

9.2 Stage 9: automatic deployment, optional

Automatic deployment updates a chosen running target and verifies it. Publishing a ZIP or image is not by itself deployment. Before this stage, choose the target, authentication model, trigger, cost controls, smoke checks, and recovery procedure [R8].

No target or paid resource is selected in this roadmap. If the learning goal is satisfied by CI and a reproducible manually run release, stage 9 can remain outside the accepted scope.

10. Operational tracking and prioritized backlog

Use a small board with To do, In progress, Review, and Done, or keep the table below in Git until a Jira board is useful. Jira setup is planned; academic Jira exposure is not being presented as an existing project board. Keep one main implementation milestone active.

ID	Task	Status	Completion evidence
DOC-01	Review/save Python automation report	Ready for review	Report + diagram assets in docs
DOC-02	Review/save scope and roadmap	Ready for review	Accepted document versions
DOC-03	Add navigation and reconcile older README/runbook status wording	Planned	Working relative links and accurate current status
DEL-01	Verify clean PBIX LFS download/open	Open check	Binary obtained, opens, reconnect behavior documented
OPS-01	Environment and driver setup	Complete	uv/Python/pyodbc and successful connection
OPS-02	Original reporting validators	Complete	18 checks and saved mismatch/recovery
OPS-03	Manifest and fingerprint baseline	Complete	743c62a, actual baseline pass
OPS-04	Five transactional loaders	Complete	Individual live first-load results
OPS-05	Safe replay and rollback	Complete	Five UNCHANGED actions, department rollback/recovery
OPS-06	Full coordinator and final validation	Complete	`bfc1763`, seven-stage PASS
CI-01	Add hosted test workflow	Planned - next	Passing automatic run
CI-02	Prove failed/corrected CI and optional required-check policy	Planned	Red/green run links and policy record
CLD-01	Define/export reproducible related subset	Planned	Manifest, counts, and local reference outputs

ID	Task	Status	Completion evidence
CLD-02	BigQuery batch demo and reconciliation	Planned	Matching subset results
DBT-01	Prove adapter/model compatibility	Conditional	Tiny model build/test succeeds
DBT-02	Build tested documented models	Planned	Model graph, tests, docs, reconciliation
ML-01	Add train-label contract/load	Planned	Counts/references/label eligibility verified
ML-02	Define candidates, cutoffs, and user partitions	Planned	Split/candidate manifest and leakage checks
ML-03	Build features and rule baseline	Planned	Reproducible feature export and baseline report
ML-04	Train/evaluate first model	Planned	Validation decisions and final holdout report
ML-05	Batch prediction and model record	Planned	Repeatable output tied to model/data versions
ORCH-01	Resolve runtime/authentication/resource feasibility	Conditional	Compatible task proof before DAG build
ORCH-02	Demonstrate scheduled task/recovery graph	Planned	Failure block and successful rerun evidence
REL-01	Release runnable accepted artifact	Conditional	Build, clean smoke run, previous-version recovery
DEP-01	Decide whether an automatic target is needed	Optional	Explicit target/scope/cost decision
COM-01	Portfolio/LinkedIn discussion	Deferred	Started separately by Saurabh

An issue should state the problem, concrete output, acceptance check, dependency, related commit, and evidence. Move it to Done only after those checks pass. Do not create a board full of tools without a result each tool is supposed to produce.

11. Decision register and compatibility gates

Decision	Resolution at this version	Revisit when
Retain validated analytics source	Keep original staging/reporting path	Explicit new-source BI integration is chosen
First CI test framework	Reuse unittest	A concrete testing need justifies another framework
CI host access	Offline checks first; no assumption of laptop access	Isolated SQL integration job is designed
BigQuery vs local fallback	Cloud subset preferred; DuckDB fallback labeled local	Access/adapter proof fails
ML size	Start with measured related customer subset	Memory/runtime supports scaling
ML target	Previously purchased next-order products	Scope expands with new candidate/evaluation design
Scheduling runtime	Separate compatible Linux/WSL2 environment	Stage 7 begins
Linux SQL connection	Must be proved; Windows auth is not assumed portable	Container/worker feasibility step
Release format	Choose working archive/image after runtime proof	Stage 8 begins
Deployment	Optional; no running target selected	A concrete use case needs automatic updates

Version pinning is stage-specific. Preserve the accepted ingestion lockfile while testing large tool dependencies in a separate environment when appropriate. Current platform limits and adapter support should be checked again when that phase starts; this roadmap is a plan, not a guarantee of future compatibility.

12. Risks, maintenance, and next handoff

Risk	Practical response
Tool collection overwhelms the project	Close one measurable milestone before adding another
Fixed baseline mistaken for daily production ingestion	Label static replay clearly; version intentional input changes
SQL and BI/new pipeline paths drift	Reconcile before changing report sources
Cloud comparisons use different data	Use the same subset manifest and reference outputs
Model target leaks into features	Record cutoff/partitions; fit only on permitted training data
Airflow/container exceeds resources	Measure small compatible setup before expansion
Published image cannot reach SQL Server	Require a real runtime connection smoke proof before release claims
Documentation becomes stale	Update statuses, evidence, links, and PDF after accepted milestones

Immediate next implementation after document review/publication: CI-01, followed by CI-02. This documentation task does not create the workflow, provision cloud resources, schedule jobs, or draft a LinkedIn post.

Maintain the Markdown files as editable sources. Regenerate the PDFs after content changes. Keep diagrams and their `.mmd` source files beside the documents. The scope states the boundaries; this roadmap states order and acceptance; detailed reports explain completed execution.

References

- R1: GitHub - Continuous integration
- R2: GitHub - Choosing a workflow runner
- R3: Google Cloud - BigQuery sandbox
- R4: dbt - Data tests
- R5: scikit-learn - Common pitfalls and data leakage
- R6: Apache Airflow - Prerequisites
- R7: Apache Airflow - Docker learning setup
- R8: GitHub - Continuous deployment
- Current implementation checkpoint
- Version history: this version supersedes the roadmap draft prepared on 29 October; earlier remote publication was not established.